BIO220 Coursework 2

Protein of Interest:

Human Angiotensin-Converting Enzyme 2 (ACE2)

ACE2 is a membrane-anchored **metalloprotease** prominently known as the **entry receptor for the SARS-CoV-2 Spike protein**. It also plays key physiological roles in the **renin–angiotensin system**, where its enzymatic activity regulates blood pressure homeostasis. Remarkably, ACE2 interacts both with another **protein partner** (the viral Spike RBD) and with **small-molecule inhibitors** that target its active site, making it an ideal model to study different modes of molecular recognition.

Marking Scheme

This assignment is worth **15% of the total module grade**. Marks are awarded for the scientific accuracy of your docking workflows, the biophysical interpretation of results, and the clarity of your structural data presentation.

Grading Criteria:

1. Explanations must demonstrate a clear understanding of molecular recognition principles and computational docking fundamentals

2. High marks are awarded for effectively linking quantitative results (e.g., PDBePISA interface areas, PLIP interaction counts, docking scores, ΔG values, RMSD) to biological interpretation, such as binding strength, interface stability, or the functional consequence of ACE2 interactions with Spike RBD or inhibitors.

3. Figures (from VMD, PyMOL, ChimeraX, or SwissDock view) must be high-resolution, labeled, and directly support the discussion. Interface residues, ligands, ions, and chains should be distinguished by color with legible annotations.

4. Reports should be well-structured and concise, integrating text, tables, and visuals logically. Captions and in-text references to figures/tables should clearly explain how each dataset supports your conclusions.

5. For exceptional grades, the work should display a critical evaluation of method limitations and propose reasoned solutions or refinements.

1. Question 1: Protein–Protein Docking (ACE2–Spike RBD Complex)

1.1 Identify and illustrate the ACE2–Spike RBD interface using the experimental complex (PDB ID: 6M0J). Isolate ACE2 (chain A) and Spike RBD (chain E) and describe the main contact region on ACE2 together with the key interacting residues. Explain briefly how you determined these residues (distance cutoff, contact analysis, or literature). Create a labelled visualization showing both proteins in cartoon representation with interface residues in sticks and distinct colors. *You may consult the original publication for guidance (doi.org/10.1038/s41586-020-2180-5)* **[15 marks]**

1.2 Reproduce the ACE2–Spike interaction using ClusPro. Upload ACE2 as the *receptor* and the Spike RBD as the *ligand* and perform docking. Record the five best docking clusters with their weighted scores and superimpose your top-ranked model onto the experimental complex. Report the Cα RMSD between the two interfaces and comment on how closely the predicted orientation matches the crystallographic structure. Identify major hydrogen bonds, salt bridges, or hydrophobic patches that stabilize your model. **[20 marks]**

1.3 Evaluate the docking model using PDBePISA. For both your docked complex and the experimental structure (6M17), use PDBePISA to calculate the buried surface area, number of hydrogen bonds, and salt bridges at the interface. Present the results in a small table and compare the trends between the two complexes. Discuss possible reasons for any differences (*hint: rigid-body approximation, side-chain flexibility, incomplete solvent model*) and conclude with a one-paragraph reflection on the reliability of ClusPro docking combined with PDBePISA analysis for studying large protein–protein interfaces. **[15 marks]**

2. Question 2: Protein–Ligand Docking of ACE2 and MLN-4760

2.1 Identify and visualize the ACE2 ligand-binding site using the experimental inhibitor-bound structure (PDB ID: 1R4L). Locate the catalytic zinc ion and describe the residues that form the MLN-4760 binding pocket. Explain briefly how these residues contribute to ligand coordination or hydrogen-bonding. Produce a labelled image showing ACE2 in cartoon representation with the inhibitor and pocket residues displayed as sticks in contrasting colors, and include a concise caption highlighting the key binding features. *You may consult the original publication for guidance (doi.org/10.1074/jbc.M311191200)* **[15 marks]**

2.2 Perform ligand docking with SwissDock to reproduce inhibitor binding. Remove MLN-4760 from the 1R4L structure to generate the apo ACE2 model. Retrieve the ligand's SMILES string from PubChem (CID 448281) and upload it to SwissDock. Set the search box around the catalytic pocket and justify your choice with a screenshot of the defined region. After docking, summarize the top five clusters with their docking scores and ΔG values (kcal/mol). Describe the key interactions in the best-scoring pose, such as hydrogen bonds, electrostatic contacts, hydrophobic contacts, or metal coordination. Compare the predicted orientation with the experimental pose in 1R4L. **[20 marks]**

2.3 Evaluate your docking results. Upload the top-ranked SwissDock complex and the experimental complex to PLIP webserver and extract the interaction profiles for each (hydrogen bonds, hydrophobic contacts, π–stacking, metal coordination, etc.). In attempting to re-dock MLN-4760 into the ACE2 active site, you may find that SwissDock does not reproduce the crystallographic binding pose. Discuss possible reasons for this failure, drawing on your understanding of docking methodology and the chemistry of the ACE2 active site (*hint: metal coordination, ligand flexibility, protonation state*). For each issue you identify, propose a realistic strategy or refinement that could improve docking accuracy. **[15 marks]**

Submission Requirements

You must submit the following files via LM. Ensure all filenames include your specific identifier (e.g., [your_name]).

For Question 1 (Protein–Protein Docking):

- **Docking Results:** Export the top five ClusPro docking clusters (in PDB format) and save your best-ranked model as `[your_name]_ClusPro_model.pdb`.

For Question 2 (Protein–Ligand Docking):

- **SwissDock Output:** Submit the docking result zip package or result directory containing all poses as `[your_name]_SwissDock_results.zip`.

Written Report:

- Submit a single PDF or Word document containing all answers, figures, tables, and discussions for both questions as `[your_name]_report.pdf`.

Final Submission Checklist

Before submitting, please verify that you have included all necessary components:

Files

- [ ] `[your_name]_ClusPro_model.pdb` (top-ranked protein–protein docking model)
- [ ] `[your_name]_SwissDock_results.zip` (all SwissDock poses and scores)
- [ ] `[your_name]_report.pdf` (main written report)

Report Content

- [ ] **Figure 1:** Labeled ACE2–Spike interface from PDB 6M0J with key residues marked.
- [ ] **Figure 2:** Table showing PDBePISA metrics (interface area and H-bond count).
- [ ] **Figure 3:** Overlay of experimental and ClusPro-predicted ACE2–Spike complexes (show structure alignment and RMSD).
- [ ] **Figure 4:** Docking search region for MLN-4760 defined around the catalytic site (Zn^{2+} shown).
- [ ] **Figure 5:** Overlay of experimental (1R4L) and SwissDock-predicted ligand poses.
- [ ] **Figure 6:** PLIP interaction diagram summarizing dominant contacts (H-bonds, hydrophobics).
- [ ] All figures have clear, descriptive captions explaining what is shown.
- [ ] All questions (1.1 to 2.3) are answered with references to figures and data where appropriate.